

0002 Definition of Node.js ECMAScript

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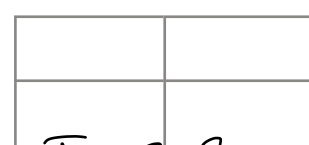
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★ Node.js is Open Source, cross-platform - JavaScript Runtime Environment.

★ ECMAScript:-

- In 1993, The first web browser with user interface called "Mosaic" was released.
- In 1994, The lead developers of 'Mosaic' founded a company called 'Netscape' & released more polished browser called 'Netscape Navigator'.
 - Web pages could only be static & there was no interactivity after a page was loaded.
- In 1995, Netscape created New Scripting language called JavaScript.
- The Name was purely for marketing purpose, as Java was the hot new language back then.
- In 1995 Microsoft also debuted their browser IE- Internet Explorer.
- Microsoft realised that JavaScript fundamentally changed the user experience of the web & wanted the same for IE.
- But there was no specification for them to follow.
- In 1996, Microsoft reverse-engineered the Navigator interpreter to create its own scripting language called JScript.
- JScript uses same use case that JavaScript, but its implementation was different.
- The differences made it difficult for developers to make their website work well in both browsers.
- In fact two implementations were so different that "Best viewed in Netscape" & "Best Viewed in IE" badges became common for most companies & couldn't afford to build for both implementation.

★ ECMA International



- In Nov 1996, Netscape submitted JavaScript to ECMA International.
- ECMA International is an Industry Association dedicated to the standardization of information & communication systems.
- Netscape wanted a standard specification that all browser vendors could conform to as it would help keep other implementations consistent across browsers.
- For each new specification ECMA provides a standard specification & a committee.
- In case of JavaScript, the standard is called ECMA-262 & the committee that works on ECMA-262 is called Technical Committee 39 (TC39).
- ECMA however decided to use the term "ECMAScript" to talk about the official language.
- The reason for this is because Oracle (who acquired Microsystems) owns the trademark for the term "JavaScript".
- So ECMAScript - refers to the standard language whereas JavaScript - is what we use in practice & builds on top of ECMAScript.

ECMAScript Versions:

- 1997 - ECMAScript 1
- 1998 - " 2
- 1999 - " 3

ECMAScript 4 - Never released.

2009 - ECMAScript 5

2015 - ECMAScript 2015 (ES6) - modern JavaScript prerequisite

one version every year since 2015.

Technically ECMAScript & JavaScript are Not the Same

ECMAScript is standard & JavaScript is based on that standard.

but these two words are often used interchangeably with each other in many book & scenarios.

ECMAScript Summary:-

- ECMA-262 is the language specification.
- ECMAScript is the language that implements ECMA-262
- JavaScript is basically ECMAScript at its core but builds on top of that.